

Centroidal MPC for Inertial Reorientation Using Bio-inspired Tailed Robot

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I. INTRODUCTION

Inertial reorientation (IR) refers to the ability of a robot to change its orientation without or with minimal interaction with the ground or other objects. This movement can increase robot agility by stabilizing the robot between ground contacts in gaits with long flight phases, rejecting sudden disturbances, or increasing the speed of the robots turns. A tail-like appendage is shown to be effective in increasing the robot's ability for inertial-reorientation [1].

However, the addition of a tail appendage makes the robot system more complex. Moreover, effective inertial reorientation control frequently necessitates control using the full-body dynamic model of the robot, as opposed to the typical approach of abstracting the robot to a single rigid body. Centroidal MPC can represent a useful compromise of these two methods [2].

II. METHODS

A. Robot Model

We use a bipedal tailed robot model. The robot has eight actuated degrees of freedom, denoted $q_j \in \mathbb{R}^8$: two for the tail, and three for each leg. The tail is modelled as two sequentially actuated joints for pitch and yaw, placed close to the torso, with a long bar and a heavy counterweight connected at the end. Each leg is modeled as three sequential joints for adduction-abduction, hip extension-flexion, and knee-extension-flexion. Additionally, the model includes six

floating base joints, $q_b \in \mathbb{R}^6$, to simulate free movement in three dimensions: three linear joints in the x, y, and z direction, and three rotating joints in the x, y, and z axis.

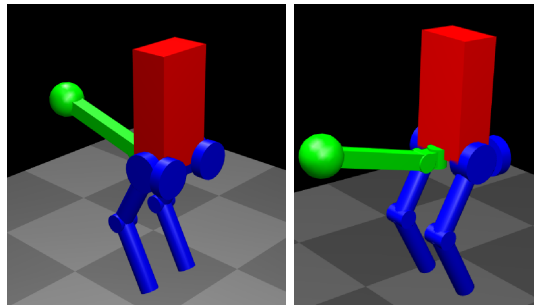


Fig. 1. Robot model with $q_j = [0 \ .25 \ 0 \ .5 \ -1 \ 0 \ .5 \ -1]^T$

We used the software package Spatial V2 [3] to model and visualize this model, shown in figure 1. The parent link, joint properties, mass, and inertia of each link are denoted in Table I, where $p(j)$ denotes the parent link of link j , Position denotes the position of the origin of the current link frame in the parent link frame, CoM Pos refers to the center of mass position in the current link frame, I refers to the diagonalized link inertia around the CoM. All link frames are constructed with identical orientation to the world frame, so frame orientations are omitted. Link 0, or the Torso, is connected to ground via six sequential floating-base joints, using the order $[Px \ Py \ Pz \ Rx \ Ry \ Rz]$. q_b and q_j are constructed with the same order as above.

TABLE I
TABLE OF ROBOT PARAMETERS.

j	Name	$p(j)$	Axis	Position	M	CoM Pos	$\text{diag}(I) \times 10^4$
0	Torso	n/a	n/a	[0 0 0]	1	[0 0 .2]	[94 83 27]
1	Tail-Y	0	z	[-0.05, 0, 0]	.05	[-0.025 0 0]	[0 0.104 0.104]
2	Tail-P	1	y	[-0.05, 0, 0]	.5	[-0.27 0 0]	[0 25.5 25.5]
3	AbAd-R	0	x	[0, -.075, 0]	.3	[0 0 0]	[0.625 22.75 22.75]
4	Hip-R	3	y	[0, 0, 0]	.75	[0 0 -0.02]	[18.25 38.8 21.8]
5	Knee-R	4	y	[0, 0, -0.2]	.1	[0 0 -0.1]	[3.33 3.33 0]
6	AbAd-L	0	x	[0, -.075, 0]	.3	[0 0 0]	[0.625 22.75 22.75]
7	Hip-L	6	y	[0, 0, 0]	.75	[0 0 -0.02]	[18.25 38.8 21.8]
8	Knee-L	7	y	[0, 0, -0.2]	.1	[0 0 -0.1]	[3.33 3.33 0]

Simulation is done using a discretized model and the `ode45` [4] function supplied with MATLAB. Specifically, we use a zero-order hold on the inputs, then simulate the system forward until the end of the time step. The state is then updated to the final state of the simulation, and the simulation repeated with more inputs generated from the centroidal MPC.

B. Centroidal Matrix Formulation

C. Trajectory Optimization Formulation

D. Centroidal MPC Formulation

III. RESULTS

IV. CONCLUSION

REFERENCES

- [1] T. Libby, A. M. Johnson, E. Chang-Siu, R. J. Full, and D. E. Koditschek, "Comparative design, scaling, and control of appendages for inertial reorientation," *IEEE Transactions on Robotics*, vol. 32, no. 6, pp. 1380–1398, 2016.
- [2] H. Dai, A. Valenzuela, and R. Tedrake, "Whole-body motion planning with centroidal dynamics and full kinematics," in *2014 IEEE-RAS International Conference on Humanoid Robots*. IEEE, 2014, pp. 295–302.
- [3] R. Featherstone, *Rigid body dynamics algorithms*. Springer, 2008.
- [4] L. F. Shampine and M. W. Reichelt, "The matlab ode suite," *SIAM journal on scientific computing*, vol. 18, no. 1, pp. 1–22, 1997.